

Week 11 Discussion Solutions

1. (a) Initially, capacitor 1 is charged while capacitor 2 has no net charge. Thus, there is a potential difference between the capacitors of $V_1 - V_2 = \frac{Q_0}{C_1}$ and current flows according to Ohm's law - $I = V/R = \frac{Q_0}{C_1 R}$.
- (b) Current flows due to the potential difference between the capacitors. As current flows and charge accumulates on capacitor 2, the potential difference diminishes and the current stops completely when the charge accumulated on capacitor 2 is such that the potential difference goes to 0, ie when $V_1 = V_2$, or

$$\frac{Q_0 - Q_2}{C_1} = \frac{Q_2}{C_2}.$$

Solving for Q_2 , we thus find that the charges on C_1 and C_2 are $\frac{C_2}{C_1+C_2}Q_0$ and $\frac{C_1}{C_1+C_2}Q_0$ respectively.

- (c) Charge accumulates onto capacitor 2 at a rate equal to the current, which itself is a function of time given by the potential difference $I(t) = V(t)/R$. Thus we have to solve the differential equation.

$$\frac{dQ}{dt} = \frac{1}{R} \left(\frac{(Q_0 - Q)}{C_1} - \frac{Q}{C_2} \right)$$

We can solve this to yield

$$Q(t) = \frac{Q_0 C_1}{C_1 + C_2} \left(e^{-\frac{(C_1+C_2)}{C_1 C_2 R} t} - 1 \right)$$

- (d) The current is the derivative of this

$$I(t) = \frac{dQ}{dt} = \frac{Q_0}{RC_2} e^{-\frac{(C_1+C_2)}{C_1 C_2 R} t} \quad (1)$$

- (e) With a dielectric in between plates of capacitor 2, the new result for $Q(t)$ is the same as in c, but with $C_2 \rightarrow \frac{\epsilon}{\epsilon_0} C_2$
2. For this problem, the capacitors act like wires at $t = 0$ (as they are uncharged), while at $t = \infty$ they behave like open circuits (as they are fully charged).
 - (a) This is the scenario for two resistors in parallel.

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2} \quad (2)$$

- (b) Across C_1 and C_2 , there is no voltage drop.

$$V_b = V, V_a = 0 \implies V_b - V_a = V \quad (3)$$

- (c) There is no current flowing through the circuit, and hence there is no voltage drop across R_1 and R_2 .

$$V_b = 0, V_a = V \implies V_b - V_a = -V \quad (4)$$

- (d) When switch 2 is closed, at equilibrium, current flows only through the resistors. R_1 and R_2 are now in series.

$$I = \frac{V}{R_1 + R_2} = \frac{V_b}{R_1} \quad (5)$$

$$V_b = V \frac{R_1}{R_1 + R_2} \quad (6)$$

- (e) Before switch 2 is closed, the charges on the two capacitors are given by

$$Q_1 = C_1 V, Q_2 = C_2 V. \quad (7)$$

After switch 2 is closed, the capacitors C_1 and C_2 are in parallel with the resistors R_2 and R_1 , respectively. The charges are given by

$$Q'_1 = C_1 (V - V_b) = C_1 V \frac{R_2}{R_1 + R_2} \quad (8)$$

$$Q'_2 = C_2 V_b = C_2 V \frac{R_1}{R_1 + R_2} \quad (9)$$

The changes in the charges are given by

$$\Delta Q_1 = Q'_1 - Q_1 = -C_1 V \frac{R_1}{R_1 + R_2} \quad (10)$$

$$\Delta Q_2 = Q'_2 - Q_2 = -C_2 V \frac{R_2}{R_1 + R_2} \quad (11)$$

3. (a) $F = i_0 dB$. It points upper left, with angle θ from the vertical.
 (b) Balancing the forces, we can write

$$\mu N = i_0 dB \sin \theta, \quad (12)$$

$$mg = N + i_0 dB \cos \theta. \quad (13)$$

Solving for B , we get

$$B = \frac{\mu mg}{i_0 d (\sin \theta + \mu \cos \theta)}. \quad (14)$$

- (c) Take a derivative of B wrt θ ,

$$\frac{\partial B}{\partial \theta} = -\frac{\mu mg}{i_0 d} \frac{\cos \theta - \mu \sin \theta}{(\sin \theta + \mu \cos \theta)^2} = 0 \quad (15)$$

has solution when $\tan \theta = \frac{1}{\mu}$. Plug this angle into the expression for B gives the smallest magnetic field.

- (d) If the magnetic field were vertical, the force would point straight left. The reason why this was not the optimal magnetic field we found in part (c) is that the wire would be sliding when it is under this force, so the analysis we did in the previous parts assuming static friction would no longer hold.
4. We will use our old friend superposition to solve this problem. Choose coordinates where the origin is at the center of the semicircle, $\hat{x}$ is to the right, and $\hat{y}$ is upwards. On the two straight edges, call the force $\vec{F}_1$.

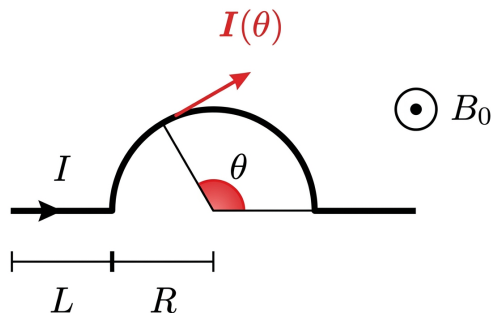
$$\vec{F}_1 = \vec{I}L \times \vec{B} = -ILB\hat{y}.$$

Keep in mind there are two such contributions from the left and right parts of the wire.

For the semi-circular part of the wire, we need to do something rather more involved. Call the force on this section of wire $\vec{F}_2$. We will parameterize the circle by θ and integrate up the forces along it using

$$d\vec{F}_2 = \vec{I}dl \times \vec{B}$$

Consider an arbitrary point on the semi-circle. See the figure below.

Figure 2: Wire showing the direction of $\vec{I}$.

We can see from the geometry that

$$d\vec{F}_2 = \vec{I}dl \times \vec{B} = (-I\hat{\theta}) \times (B\hat{z}) dl = -IBRd\theta\hat{s}$$

Since the force points inwards all along the circle (the $-\hat{s}$ direction), we can see that the horizontal components will cancel out, and the vertical components will add. Therefore, the net force will be downwards, and we need only consider

$$dF_y = \hat{y} \cdot d\vec{F}_2 = -IBRd\theta (\hat{y} \cdot \hat{s}) = -IBR \sin\theta d\theta$$

Now, we need to integrate from $\theta = 0$ to $\theta = \pi$. The fact that current flows from $\theta = \pi$ to $\theta = 0$ does not matter; when we computed $d\vec{F}_2$, we assumed that $dl > 0$. If we integrate from $\theta = \pi$ to $\theta = 0$, it has the effect that $d\theta < 0$, which would correspond to a magnetic force pointing outward along the circle, but we know that the force points inward in this case.

$$\vec{F}_2 = F_y\hat{y} = \int_0^\pi -IBR\hat{y} \sin\theta d\theta = +IBR\hat{y} \cos\theta \Big|_0^\pi = -2IBR\hat{y}.$$

Adding the contributions from the straight sections of wire,

$$\vec{F} = -ILB\hat{y} - 2IBR\hat{y} - ILB\hat{y} = -I(2L + 2R)B\hat{y}. \quad (16)$$

It is interesting to note that the semi-circle completely dropped out — it looks like it was just a straight wire of length $2L + 2R$. Why was this? Can you show that only the x -length matters for a wire with an arbitrary curve in the same constant magnetic field?